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A Quantile-Adaptive Profiled-Score Mixed-Effects Model for ICU Blood Pressure Trajectories

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Abstract

Background: Intensive care unit (ICU) monitoring data are longitudinal, irregularly sampled, and clinically heterogeneous. For hemodynamic variables such as mean arterial pressure (MAP), clinically important deterioration is often concentrated in the lower tail rather than in the conditional mean. Mean-based trajectory models may therefore obscure persistent hypotension burden and delayed hemodynamic recovery. **Methods:** We develop a quantile-adaptive profiled-score mixed-effects representation for irregular longitudinal trajectories. The model combines a flexible reference conditional quantile trajectory with stay-specific low-rank deviations learned directly under a smooth quantile objective. To close identification, we separate the intercept from the time-varying trajectory through a centered spline dictionary and impose an empirical centering constraint on the stay-specific score vectors. Estimation is carried out by blockwise optimization with manifold-constrained loading updates and stay-level cross-validation based on within-stay forecasting. **Results:** The simulation study is designed to evaluate recovery of lower-tail trajectories, recovery of the heterogeneous low-rank subspace, empirical bootstrap interval behavior, and numerical stability under irregular sampling, heavy-tailed noise, and informative-observation sensitivity scenarios. The MIMIC-IV analysis will report cohort construction, including `[N_stays]` adult ICU stays and `[N_MAP]` MAP measurements after finalized analyses, within-stay forecasting performance, threshold-based hypotension summaries, and stability of the estimated lower-tail deviation subspace. **Conclusions:** The proposed framework is intended to provide an interpretable and computationally feasible approach for lower-tail longitudinal modeling in ICU monitoring studies. Its main contribution is to learn the low-rank trajectory representation under the quantile target itself rather than importing a mean-oriented basis into a downstream quantile analysis.

KEYWORDS

quantile regression, profiled-score mixed-effects representation, reduced-rank modeling, trajectory heterogeneity, intensive care unit, mean arterial pressure, irregular longitudinal data, MIMIC-IV

1 | INTRODUCTION

Intensive care unit (ICU) monitoring data are inherently longitudinal, irregularly sampled, and clinically heterogeneous. Physiologic variables are recorded repeatedly over time, but the timing and frequency of measurement are determined by clinical workflow and patient acuity rather than by a prespecified research protocol. As a result, ICU trajectories are noisy, unevenly observed, and often poorly summarized by models that focus exclusively on the conditional mean. Publicly available critical care resources such as MIMIC-IV provide an important opportunity to develop and evaluate statistical methods under these realistic conditions[1,2]. In the present paper, we focus on mean arterial pressure (MAP), a routinely monitored hemodynamic variable whose lower-tail behavior is directly relevant to critical care practice[3,4].

Mean-based longitudinal models are often insufficient when the scientific interest lies in clinically meaningful risk regions rather than in average trajectory behavior. For MAP, transient or sustained low-pressure episodes may indicate inadequate perfusion or delayed hemodynamic stabilization even when the average trajectory remains relatively unremarkable. This motivates a distribution-sensitive perspective in which lower conditional quantiles of the trajectory are treated as primary estimands. Since the introduction of regression quantiles by Koenker and Bassett[5], quantile regression has provided a principled framework for modeling different parts of the conditional distribution and has become a robust alternative to least-squares regression when tail behavior or heterogeneous covariate effects are of interest. Extensions to longitudinal and clustered settings further established the relevance of quantile methods for repeated-measures data[6,7,8,9].

For dependent data, mixed-effects quantile models provide a natural extension of quantile regression by introducing latent cluster-specific effects to account for within-cluster dependence[6,7,8,9]. Existing work has shown that this framework can be effective for longitudinal data and can be implemented through either likelihood-based or optimization-based strategies[7,8,9,10]. However, most mixed-effects quantile formulations remain conceptually close to classical random intercept and random slope models. Such structures are often too restrictive for ICU monitoring data, where the most salient differences across stays concern trajectory shape rather than only overall level or linear trend. In practice, some stays exhibit persistently depressed MAP trajectories, others show delayed recovery, and others display substantial instability in lower-tail behavior. Capturing these patterns requires a trajectory-level representation of heterogeneity rather than only a small number of scalar random coefficients tied to simple time effects.

A common way to represent trajectory heterogeneity is through reduced-rank or functional modeling. In sparse longitudinal settings, functional principal component analysis (FPCA) provides a well-established low-dimensional representation of subject-specific curves, while functional mixed-effects models and reduced-rank mixed models allow complex trajectory variation to be summarized through a small number of latent directions and associated scores[11,12,13,14]. These ideas are highly attractive in ICU monitoring because they can accommodate irregular observation times and preserve interpretable low-dimensional summaries of trajectory variation. Nevertheless, standard reduced-rank trajectory methods are fundamentally driven by mean and covariance structure. They identify directions of dominant overall variability rather than directions that are specifically informative for a target conditional quantile.

This distinction becomes important when the inferential target is a lower-tail MAP trajectory. The directions that best explain average between-stay variation need not coincide with those that best explain clinically relevant lower-tail dynamics. A mean-oriented FPCA basis may therefore provide an inadequate representation when the primary scientific goal is to characterize sustained hypotension risk or delayed recovery in the lower conditional quantiles. Functional quantile regression has developed substantially in recent years, demonstrating that quantile structure itself can serve as the basis for trajectory-level modeling rather than being treated as an afterthought to mean-based dimension reduction[15,16]. More recently, functional quantile principal component analysis has explicitly shown that dimension reduction can be adapted to participant-specific quantile curves rather than restricted to classical mean-oriented functional principal components[17]. These developments suggest that lower-tail trajectory analysis should not automatically rely on a basis learned under a mean-based objective.

At the same time, we do not claim that quantile-adapted basis learning is new in itself. The narrower goal of this paper is to embed quantile-adaptive low-rank trajectory learning in an irregular longitudinal setting and to use profiled stay-level scores directly for conditional lower-tail trajectory modeling. The incremental value of doing so must be evaluated through simulation and through the MIMIC-IV application, especially by comparison with a two-stage FPCA-plus-quantile strategy that learns a mean-oriented basis before fitting the lower-tail model. We therefore position the contribution as an applied methodological integration rather than as a claim of first discovery for quantile-adaptive dimension reduction.

Citation TODO: before final submission, the literature review should be checked and supplemented for convolution-smoothed or kernel-smoothed quantile regression, functional quantile regression for sparse or irregular data, quantile dimension reduction and quantile sufficient dimension reduction, longitudinal reduced-rank functional models, informative observation processes in longitudinal data, and ICU MAP, hypotension burden, acute kidney injury, mortality, and length-of-stay studies. Existing bibliography keys should be used where available; otherwise the relevant entries should be added deliberately rather than guessed.

Our motivating clinical question is descriptive and prognostic rather than causal: among adult ICU stays during the first 24 hours after admission, how does the lower conditional trajectory of MAP evolve over time after adjustment for baseline case-mix, and what low-dimensional forms of between-stay heterogeneity characterize persistently low or delayed-recovery trajectories? The intended use of the model is therefore twofold. First, it provides a cohort-level summary of lower-tail hemodynamic behavior during early ICU time. Second, once partial trajectory information is available for a stay, it yields an interpretable low-rank decomposition of stay-specific lower-tail deviations.

This distinction is important because the target of the present model is a conditional quantile trajectory, not the probability that MAP falls below 65 mmHg at a given time. We focus on $\tau = 0.10$ because it places estimation weight on clinically relevant lower-tail behavior while remaining numerically stable under irregular monitoring. To connect this distributional target back to bedside practice, the empirical section also reports a threshold-based supplementary summary based on the observed proportion of time with MAP below 65 mmHg across strata of the estimated lower-tail score. In this way, the model addresses a clinically meaningful low-pressure problem without conflating a conditional quantile estimand with a threshold-exceedance probability.

The present paper is motivated by this gap. We consider the problem of modeling lower-tail MAP trajectories during early ICU time and argue that the trajectory representation itself should be aligned with the target conditional quantile. This is particularly relevant in critical care because current sepsis and septic shock guidelines recommend an initial MAP target around 65 mmHg when vasopressors are required, underscoring the clinical salience of lower-tail blood pressure behavior [3,18]. Moreover, recent systematic evidence indicates that exposure to hypotension during ICU stay is associated with increased mortality and acute kidney injury, and that these associations tend to strengthen as hypotension severity increases [4]. These considerations motivate a statistical framework that targets lower-tail trajectory dynamics directly rather than relying on a mean-based surrogate.

To address this problem, we develop a quantile-adaptive profiled-score mixed-effects representation for irregular ICU trajectories. The key idea is to replace the usual two-stage strategy–first learning a reduced-rank trajectory basis under a mean-oriented criterion and then fitting a downstream quantile model–with a joint formulation in which the low-rank trajectory representation is estimated directly under the target quantile objective. The resulting representation is therefore adapted to lower-tail dynamics rather than to global covariance-dominated variation. In the working representation, the reference conditional quantile trajectory is represented flexibly over time, while stay-specific deviations are summarized through a small number of quantile-adaptive basis functions and profiled stay-specific scores. This construction preserves the interpretability and computational advantages of reduced-rank longitudinal modeling while aligning trajectory representation with the actual inferential target.

The proposed method differs from classical longitudinal quantile mixed models because the heterogeneous component is not restricted to prespecified random intercepts and slopes. It also differs from two-stage FPCA-plus-quantile procedures, which first estimate a trajectory basis under mean and covariance criteria and then treat that basis as fixed in a downstream quantile model. Finally, it differs from functional quantile principal component approaches because our goal is a conditional longitudinal quantile model with stay-specific profiled scores under irregular observation times, rather than quantile-oriented dimension reduction of subject-level curves estimated in a preliminary step. Our contribution is therefore the joint estimation of a low-rank trajectory subspace and a conditional quantile model within a single irregular longitudinal framework.

Our motivating application uses MIMIC-IV, a large and publicly available ICU electronic health record dataset that contains time-stamped physiologic measurements, treatments, diagnoses, and other clinical variables [1,2]. The modular structure of MIMIC-IV and the availability of ICU event tables make it particularly suitable for stay-level longitudinal analysis. At the same time, the dataset retains the irregularity and noise characteristic of routine ICU documentation, making it a realistic environment for assessing whether the proposed model can deliver an interpretable and stable lower-tail trajectory representation under practical data conditions [1,2].

The contributions of this paper are fourfold. First, we formulate a quantile-specific reduced-rank trajectory model whose reference trajectory is identified through a centered spline dictionary and centered stay-level scores, while the heterogeneous component is interpreted through a rotation-invariant low-rank subspace. Second, we develop a practical estimation strategy using a smooth quantile surrogate, blockwise optimization, and manifold-constrained loading updates. Third, we adopt a validation scheme based on within-stay forecasting rather than naive held-out clustering, thereby aligning model selection with the intended use of the heterogeneous component. Fourth, we provide a deliberately modest theoretical justification targeted at the estimator actually implemented in practice and evaluate the method through simulation and a MIMIC-IV application focused on lower-tail MAP trajectories.

The rest of the paper is organized as follows. Section 2 introduces the proposed model and its identification. Section 3 describes estimation, tuning selection, and pragmatic uncertainty summaries. Section 4 presents a limited large-sample justification for the implemented estimator. Section 5 reports the simulation design and the finite-sample outputs to be populated from the finalized runs. Section 6 presents the MIMIC-IV application. Section 7 concludes with a discussion.

2 | QUANTILE-ADAPTIVE PROFILED-SCORE MIXED-EFFECTS REPRESENTATION

2.1 | Setting and notation

Let $i = 1, \dots, N$ index ICU stays and $j = 1, \dots, n_i$ index repeated measurements within stay i. For stay i, let Y_{ij} denote the observed response at time t_{ij}, measured in hours since ICU admission. In the motivating application, Y_{ij} is MAP and the analysis window covers the first 24 hours after admission. Both the number of observations n_i and the observation times $\{t_{ij}\}_{j=1}^{n_i}$ vary freely across stays, reflecting the irregular sampling that characterizes routine ICU monitoring.

Let $x_i^\circ \in \mathbb{R}^{p-1}$ denote a vector of baseline covariates with the intercept removed. The intercept is treated separately in the model below so that it cannot be confounded with the time-varying component. For a fixed quantile level $\tau \in (0,1)$, the objective is to model the conditional τ-quantile trajectory of Y_{ij} over time while allowing for stay-specific departures from the reference trajectory. The observed data are

$$\left\{Y_{ij}, t_{ij}, x_i^\circ : j = 1, \dots, n_i,\ i = 1, \dots, N\right\},$$

and we write $M_N = \sum_{i=1}^N n_i$ for the total number of observations.

Temporal structure is represented through a pre-specified dictionary of smooth basis functions

$$B^\star(t) = \left(b_1^\star(t), \dots, b_L^\star(t)\right)^\top,$$

defined on the observation window $[0, T]$. To separate the intercept from the time-varying component, we work with the centered dictionary

$$\widetilde{B}(t) = B^\star(t) - \bar{B}, \qquad \bar{B} = T^{-1} \int_0^T B^\star(u)\, du,$$

so that

$$\int_0^T \widetilde{B}(t)\, dt = 0.$$

Let

$$G_{\widetilde{B}} = \int_0^T \widetilde{B}(t)\widetilde{B}(t)^\top\, dt$$

denote the Gram matrix of the centered basis system. For later use, define the admissible loading set

$$\mathcal{C}_K = \{C \in \mathbb{R}^{L\times K} : C^\top G_{\widetilde{B}} C = I_K\}.$$

2.2 | Model formulation and identification

The following formulation should be understood as a working profiled quantile representation rather than as a fully parametric random-effects likelihood model. The stay-specific score a_i is introduced to summarize low-dimensional trajectory heterogeneity and is estimated by profiling; it is not integrated out over a specified mixing distribution. Consequently, the primary population-level target is the reference conditional quantile trajectory and the low-rank deviation subspace induced by the profiled criterion, rather than the distribution of individual random effects.

For a given quantile level $\tau \in (0,1)$, we use the working profiled representation

$$Q_\tau(Y_{ij} \mid x_i^\circ, t_{ij}, a_i) = \alpha_\tau + x_i^{\circ\top}\beta_\tau + \widetilde{B}(t_{ij})^\top\theta_\tau + \widetilde{B}(t_{ij})^\top C_\tau a_i, \tag{1}$$

where $\alpha_\tau \in \mathbb{R}$ is a scalar intercept, $\beta_\tau \in \mathbb{R}^{p-1}$ is a vector of fixed-effect coefficients, $\theta_\tau \in \mathbb{R}^L$ describes the common time-varying quantile component, $C_\tau \in \mathbb{R}^{L\times K}$ is a loading matrix whose columns define K quantile-adaptive deviation directions, and $a_i \in \mathbb{R}^K$ is the stay-specific score vector. The rank K is assumed to be small relative to L, so that between-stay trajectory heterogeneity is summarized parsimoniously. We write

$$\vartheta_\tau = (\alpha_\tau, \beta_\tau^\top, \theta_\tau^\top)^\top$$

for the common parameter block.

Writing

$$C_\tau = (c_{1,\tau}, \dots, c_{K,\tau}),$$

with $c_{k,\tau} \in \mathbb{R}^L$ denoting the kth column of C_τ, define

$$\psi_{k,\tau}(t) = \widetilde{B}(t)^\top c_{k,\tau}, \qquad k = 1, \dots, K.$$

Then (1) can be rewritten as

$$Q_\tau(Y_{ij} \mid x_i^\circ, t_{ij}, a_i) = \alpha_\tau + x_i^{\circ\top}\beta_\tau + \mu_\tau^{\mathrm{ref}}(t_{ij}) + \sum_{k=1}^{K} a_{ik}\psi_{k,\tau}(t_{ij}), \tag{2}$$

where

$$\mu_\tau^{\mathrm{ref}}(t) = \widetilde{B}(t)^\top\theta_\tau, \qquad q_\tau^{\mathrm{ref}}(t) = \alpha_\tau + \mu_\tau^{\mathrm{ref}}(t).$$

Here $\mu_\tau^{\mathrm{ref}}(t)$ is the centered time-varying component of the common quantile trajectory, while $q_\tau^{\mathrm{ref}}(t)$ is the *reference conditional τ-quantile trajectory*: it corresponds to a stay with centered baseline covariates $x_i^\circ = 0$ and centered latent score $a_i = 0$. It is not a marginal population quantile curve. This distinction is important both statistically and clinically, because the model is conditional on the included baseline covariates and on the stay-specific heterogeneous component.

The product $C_\tau a_i$ is invariant to reparameterization. To remove the resulting scale ambiguity and to close the decomposition between the common time-varying component and the stay-specific deviations, we impose

$$C_\tau^\top G_{\widetilde{B}} C_\tau = I_K, \qquad \frac{1}{N}\sum_{i=1}^{N} a_i = 0. \tag{3}$$

The first condition fixes the scale of the loading matrix in the $G_{\widetilde{B}}$ metric. The second removes translational non-identifiability between θ_τ and the score vectors. Indeed, without the empirical centering constraint, the decomposition would remain unchanged under the transformation

$$\theta_\tau \leftarrow \theta_\tau + C_\tau b, \qquad a_i \leftarrow a_i - b,$$

for arbitrary $b \in \mathbb{R}^K$.

The orthonormality constraint does not remove orthogonal rotational freedom: if O is any $K \times K$ orthogonal matrix, then $(C_\tau O, O^\top a_i)$ satisfies (3) and induces the same fitted trajectories. Consequently, the individual columns of C_τ are not point-identified, but the K-dimensional column space of C_τ and the induced stay-specific deviation process $\widetilde{B}(t)^\top C_\tau a_i$ are uniquely determined. Since the heterogeneous trajectory component rather than a particular rotated representative is the main object of substantive interest, this level of identification is sufficient for the purposes of the paper.

The stay-specific score vectors are treated as profiled nuisance parameters rather than integrated out over a parametric mixing distribution. We therefore do not interpret the procedure as likelihood-based random-effects inference. Between-stay heterogeneity refers here to heterogeneity captured by these profiled stay-level scores within the low-rank trajectory space. This choice avoids strong distributional assumptions, but it also implies that the method is best viewed as a descriptive and partially predictive framework whose stay-specific updating becomes meaningful only after partial trajectory information is available for a given stay. In particular, with bounded or modest within-stay observation counts, the theory below concerns the common component and the low-rank subspace of the profiled population criterion; it should not be read as a claim that individual latent scores are consistently recovered as structural random effects.

The rank K is a structural parameter governing how many independent directions of trajectory heterogeneity are modeled. Setting $K = 0$ recovers a standard longitudinal quantile regression with no stay-specific trajectory component. Small positive K with a very low-dimensional basis approximates conventional random-intercept and random-slope quantile models, while the general case with $K \geq 2$ and a richer spline basis captures more complex patterns such as sustained lower-tail suppression or delayed recovery.

For any admissible loading matrix C, write $\mathcal{S}(C)$ for its K-dimensional column space under the $G_{\widetilde{B}}$ inner product. For two admissible loading matrices C_1 and C_2, let $\phi_1, \ldots, \phi_K$ denote the corresponding principal angles, equivalently defined through the singular values of $C_1^\top G_{\widetilde{B}} C_2$. We use the rotation-invariant subspace distance

$$d\big(\mathcal{S}(C_1), \mathcal{S}(C_2)\big) = \left\{K^{-1}\sum_{k=1}^{K}\sin^2(\phi_k)\right\}^{1/2}.$$

Proposition 1 (Identification up to rotation)**.** *Under the centering and scale constraints in* (3) *and the design-separation conditions in Assumption 2, the reference parameter block $(\alpha_\tau, \beta_\tau, \theta_\tau)$ is identified within the working profiled representation. The loading matrix C_τ is identified only up to post-multiplication by an orthogonal matrix. Consequently, the identifiable heterogeneous component is the low-rank subspace $\mathcal{S}(C_\tau)$ and rotation-invariant functionals of the induced deviation process, rather than a particular orientation of the loading columns or structurally identified individual score vectors.*

2.3 | Regularity conditions

The following assumptions are stated at the level of generality appropriate for the fixed-dimensional irregular longitudinal setting considered here. They are formulated for the estimator actually implemented in practice rather than for an unrealistically broad class of asymptotic regimes.

Assumption 1 (Independent stays and bounded within-stay information)**.** The stay-level observations $W_1, \dots, W_N$ are independent and identically distributed, where

$$W_i = \{Y_{ij}, t_{ij}, x_i^\circ : j = 1, \dots, n_i\}.$$

There is a finite constant $\bar{n}$ such that $1 \leq n_i \leq \bar{n} < \infty$ almost surely, and all observation times lie in $[0, T]$. Observation times are treated as conditionally exogenous for the working conditional quantile target given the variables included in the model. Because this assumption may be restrictive in ICU monitoring data, its empirical impact is assessed by sensitivity analyses.

Assumption 2 (Compactness, bounded design, and basis regularity)**.** The common parameter block satisfies $\vartheta_\tau \in \Theta$, where Θ is compact, and the loading matrix is restricted to $\mathcal{C}_K = \{C : C^\top G_{\widetilde{B}} C = I_K\}$. The centered basis functions in $\widetilde{B}(t)$ are linearly independent, contain no constant component, and are uniformly bounded on $[0, T]$. The Gram matrix

$$G_{\widetilde{B}} = \int_0^T \widetilde{B}(t)\widetilde{B}(t)^\top dt$$

is positive definite. Baseline covariates are assumed bounded, standardized, or truncated for the purpose of the theoretical argument, and $E|Y_{ij}| < \infty$ for each within-stay index $j \leq \bar{n}$.

The following design-separation conditions are also imposed on the working representation. The observation-time distribution has support rich enough to distinguish all centered basis directions on $[0, T]$, and the joint support of (x_i°, t_{ij}) is rich enough that

$$\delta_\alpha + x_i^{\circ\top} \delta_\beta + \widetilde{B}(t_{ij})^\top \delta_\theta = 0$$

almost surely implies $\delta_\alpha = 0$, $\delta_\beta = 0$, and $\delta_\theta = 0$. Finally, the common fixed-effect component is separated from the admissible low-rank deviation component: for any admissible C satisfying $C^\top G_{\widetilde{B}} C = I_K$, if

$$\delta_\alpha + x_i^{\circ\top} \delta_\beta + \widetilde{B}(t_{ij})^\top \delta_\theta = \widetilde{B}(t_{ij})^\top C b_i$$

almost surely for some stay-level vectors b_i satisfying the same centering convention as the scores, then $\delta_\alpha = 0$, $\delta_\beta = 0$, $\delta_\theta = 0$, and $\widetilde{B}(t_{ij})^\top C b_i = 0$ almost surely. The dimensions of the common parameter block and the working rank K are fixed, with $K < L$.

Remark 1. The design-separation part of Assumption 2 is a technical identification condition rather than a primitive condition that can be verified directly from the observed data. It rules out cases in which the common trajectory component and the low-rank deviation component are observationally indistinguishable. In a finite-dimensional spline dictionary with rich observation-time support, the centered score convention, and sufficient variation in baseline covariates and time basis functions, this is the standard type of design-separation requirement needed for a reduced-rank longitudinal representation. It should not be read as a claim of global identification of individual score vectors.

Assumption 3 (Profiled score regularity)**.** For fixed $h > 0$ and $\lambda_a > 0$, the inner minimization over $a \in \mathbb{R}^K$ in the profiled stay-level loss exists for every $(\vartheta_\tau, C_\tau) \in \Theta \times \mathcal{C}_K$. A measurable minimizer can be selected. The profiled stay-level loss is measurable in W_i and is almost surely continuous in (ϑ_τ, C_τ).

Assumption 4 (Local separation)**.** The population profiled criterion has a locally separated minimizer $(\vartheta^\circ_{\tau,h,\lambda}, \mathcal{S}^\circ_{\tau,h,\lambda})$ in the metric

$$\|\vartheta - \vartheta^\circ_{\tau,h,\lambda}\|_2 + d\{\mathcal{S}(C), \mathcal{S}^\circ_{\tau,h,\lambda}\}.$$

That is, for every sufficiently small neighborhood of the pseudo-true parameter-subspace pair, the population profiled criterion is strictly larger on the boundary of that neighborhood than at the pseudo-true pair.

Together, Assumptions 1–4 characterize a fixed-dimensional irregular longitudinal setting. They do not assert global identification or global convexity. The primary inferential challenge addressed here is quantile-oriented low-rank trajectory learning under irregular observation times and between-stay heterogeneity, rather than high-dimensional ambient parameter spaces.

3 | ESTIMATION AND PRAGMATIC INFERENCE

3.1 | Penalized estimation under a smooth quantile objective

A natural starting point for estimation is the quantile check loss

$$\rho_\tau(u) = u\{\tau - \mathbb{I}(u < 0)\}.$$

However, direct optimization of the nonsmooth check loss becomes unstable once the common parameter block, the loading matrix, and the stay-specific score vectors are estimated jointly. The low-rank term $\widetilde{B}(t_{ij})^\top C_\tau a_i$ makes the overall criterion nonconvex, and the nondifferentiability of the check loss makes blockwise updating less stable than in ordinary smooth estimation problems. For this reason, we work with the differentiable surrogate

$$\rho_{\tau,h}(u) = \tau u + h \log\{1 + \exp(-u/h)\}, \tag{4}$$

where $h > 0$ is a smoothing bandwidth. Its derivative is

$$\psi_{\tau,h}(u) = \frac{\partial}{\partial u}\rho_{\tau,h}(u) = \tau - \{1 + \exp(u/h)\}^{-1}. \tag{5}$$

As $h \downarrow 0$, $\rho_{\tau,h}(u)$ converges pointwise to the check loss. The working representation in Section 2 motivates the scientific estimand, but the estimator analyzed in Section 4 targets the pseudo-true minimizer of the implemented smoothed and penalized criterion for fixed tuning values. Statements about recovery of the unsmoothed structural conditional quantile target would require additional conditions on tuning sequences, such as $h \downarrow 0$ and suitably vanishing regularization; those stronger claims are not made here.

For fixed h, the estimator targets the minimizer of the smoothed loss rather than the exact check-loss criterion. The resulting smoothing bias depends on the local conditional density of the residual around zero and on the asymmetry of the conditional distribution; its direction is not assumed a priori. The empirical analysis therefore reports sensitivity to h, and formal recovery of the unsmoothed target would require additional assumptions on a vanishing bandwidth sequence.

Using the common block notation above, the residual is

$$r_{ij}(\vartheta_\tau, C_\tau, a_i) = Y_{ij} - \alpha_\tau - x_i^{\circ\top}\beta_\tau - \widetilde{B}(t_{ij})^\top\theta_\tau - \widetilde{B}(t_{ij})^\top C_\tau a_i. \tag{6}$$

We estimate

$$\left\{\alpha_\tau, \beta_\tau, \theta_\tau, C_\tau, \{a_i\}_{i=1}^N\right\}$$

by minimizing

$$\mathcal{Q}_{N,\tau,h,\lambda}(\vartheta_\tau, C_\tau, \{a_i\}_{i=1}^N) = \frac{1}{N}\sum_{i=1}^N \left[\sum_{j=1}^{n_i} \rho_{\tau,h}\left\{r_{ij}(\vartheta_\tau, C_\tau, a_i)\right\} + \lambda_a\|a_i\|_2^2\right] + \lambda_R \mathcal{R}(C_\tau), \tag{7}$$

subject to (3), where

$$\mathcal{R}(C_\tau) = \sum_{k=1}^K \|Dc_{k,\tau}\|_2^2 = \mathrm{tr}(C_\tau^\top D^\top D C_\tau),$$

and D is the second-difference matrix applied to spline coefficients. The penalty on a_i stabilizes the heterogeneous component for sparsely observed stays, while $\mathcal{R}(C_\tau)$ discourages excessively oscillatory loading functions without altering the rotation-invariant interpretation of the estimated subspace. The stay-level normalization makes the sample criterion correspond directly to the population profiled criterion in Section 4. Equivalently, one may minimize the unnormalized criterion obtained by multiplying (7) by N, in which case the loading penalty must be written as $N\lambda_R\mathcal{R}(C_\tau)$.

3.2 | Blockwise estimation

The objective (7) is not jointly convex, but it decomposes into lower-dimensional subproblems that can be solved in alternation. We therefore use a blockwise procedure that cycles among the common parameter block $(\alpha_\tau, \beta_\tau, \theta_\tau)$, the stay-specific score vectors $\{a_i\}_{i=1}^N$, and the loading matrix C_τ, with each block updated by a smooth gradient-based method under the differentiable loss (4).

Initialization proceeds in three steps. First, an initial estimate of $(\alpha_\tau, \beta_\tau, \theta_\tau)$ is obtained by fitting the marginal smoothed quantile model with the heterogeneous term omitted. Second, for each stay a ridge-stabilized basis coefficient vector is computed from the residuals, centered across stays, and assembled into a coefficient matrix. Third, a rank-K singular value decomposition of the whitened coefficient matrix produces an initial loading matrix and initial score vectors. Because the problem is nonconvex, all reported fits use one structured initialization together with 20 random perturbation starts; the feasible solution with the smallest final objective value is retained.

Given current values of C_τ and $\{a_i\}$, the first update step minimizes (7) with respect to $\vartheta_\tau = (\alpha_\tau, \beta_\tau^\top, \theta_\tau^\top)^\top$. In this step the heterogeneous component is treated as an offset, so the problem reduces to a finite-dimensional smooth quantile regression. The corresponding first-order condition can be written compactly as

$$\sum_{i=1}^{N} Z_i^\top \psi_{\tau,h}(r_i) = 0,$$

where Z_i is the stay-level design matrix and $\psi_{\tau,h}(r_i)$ is applied componentwise. This block is updated by a limited-memory quasi-Newton method.

Given $(\alpha_\tau, \beta_\tau, \theta_\tau)$ and C_τ, the second update step minimizes (7) with respect to the score vector a_i separately for each stay. Because K is small, each of these score updates is low-dimensional. The stay-specific updates are conditionally independent given the other parameters and are therefore naturally parallelizable.

Given $(\alpha_\tau, \beta_\tau, \theta_\tau)$ and the updated score vectors, the third step takes a smooth gradient-based update in the loading matrix using the penalized criterion

$$\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{n_i} \rho_{\tau,h}\big(r_{ij}(\vartheta_\tau, C_\tau, a_i)\big) + \lambda_R \mathcal{R}(C_\tau),$$

followed by retraction onto the constraint set

$$C_\tau^\top G_{\widetilde{B}} C_\tau = I_K.$$

The score penalty is omitted from this display because the score vectors are fixed during the loading update, so $\lambda_a \|a_i\|_2^2$ is constant with respect to C_τ. This is the step that makes the estimator quantile-adaptive, since the low-rank trajectory directions are updated under the quantile fitting criterion rather than imported from an external mean-based decomposition.

For implementation, let

$$z_{ij} = \Big(1,\ x_i^{\circ\top},\ \widetilde{B}(t_{ij})^\top\Big)^\top$$

and let $r_{ij} = r_{ij}(\vartheta_\tau, C_\tau, a_i)$. Under the normalized criterion (7), the gradient with respect to the common block is

$$\nabla_\vartheta \mathcal{Q}_{N,\tau,h,\lambda} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{n_i} z_{ij}\psi_{\tau,h}(r_{ij}). \tag{8}$$

For the stay-specific score a_i, ignoring the common factor N^{-1}, the gradient is

$$\nabla_{a_i} = -\sum_{j=1}^{n_i} C_\tau^\top \widetilde{B}(t_{ij})\psi_{\tau,h}(r_{ij}) + 2\lambda_a a_i. \tag{9}$$

The Euclidean gradient with respect to C_τ is

$$\nabla_C \mathcal{Q}_{N,\tau,h,\lambda} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{n_i} \widetilde{B}(t_{ij}) a_i^\top \psi_{\tau,h}(r_{ij}) + 2\lambda_R D^\top D C_\tau. \tag{10}$$

After a tentative loading update $\widetilde{C}$, feasibility is restored by the generalized Stiefel retraction

$$\mathcal{R}_G(\widetilde{C}) = \widetilde{C}\left(\widetilde{C}^\top G_{\widetilde{B}} \widetilde{C}\right)^{-1/2}, \tag{11}$$

provided that $\widetilde{C}^\top G_{\widetilde{B}} \widetilde{C}$ is nonsingular. This guarantees

$$\mathcal{R}_G(\widetilde{C})^\top G_{\widetilde{B}} \mathcal{R}_G(\widetilde{C}) = I_K.$$

After each full sweep of the algorithm, the score vectors are recentered to satisfy the identifying constraint:

$$\bar{a} = N^{-1}\sum_{i=1}^{N} a_i, \qquad \theta_\tau \leftarrow \theta_\tau + C_\tau \bar{a}, \qquad a_i \leftarrow a_i - \bar{a}, \quad i = 1, \ldots, N.$$

This transformation leaves all fitted trajectories unchanged and enforces the identifying convention. Moreover,

$$\sum_{i=1}^{N} \|a_i - \bar{a}\|_2^2 = \sum_{i=1}^{N} \|a_i\|_2^2 - N\|\bar{a}\|_2^2.$$

so the centered representation weakly decreases the ridge penalty on the score vectors. Hence any noncentered representation is dominated, in the penalized criterion, by its centered equivalent after the corresponding update of θ_τ. Because the loading matrix has already been retracted in the third block update, the orthonormality constraint is preserved at this stage.

Convergence is monitored through the relative change in the objective function and the maximum change across parameter blocks. Because the criterion is nonconvex, convergence to a global minimizer cannot be guaranteed. For this reason, the simulation study reports convergence frequency, dispersion of final objective values across starts, and stability of the estimated subspace under alternative initializations.

3.3 | Choice of tuning parameters

The estimator involves the smoothing parameter h, the rank K, and the regularization parameters (λ_a, λ_R). Smaller h brings the criterion closer to the unsmoothed quantile loss but may slow convergence; larger h improves smoothness at the cost of greater approximation bias. The rank K governs how many independent directions of lower-tail trajectory heterogeneity are represented. The parameter λ_a shrinks stay-specific score vectors toward zero, providing stability for stays with sparse or uneven observation times. The smoothness penalty parameter λ_R controls the roughness of the loading functions.

All tuning quantities are selected jointly by stay-level cross-validation combined with within-stay forecasting. In each fold, the common parameters $(\alpha_\tau, \beta_\tau, \theta_\tau, C_\tau)$ are estimated on the training stays only. For each held-out stay, the stay-specific score a_i is then estimated using only the earliest 50% of its observations (or all observations before 12 hours when sufficiently many measurements are available), while validation performance is evaluated by the *unsmoothed* check loss on the remaining observations of that stay. This design avoids using the full held-out trajectory to estimate a_i and better matches the intended use of the model, namely lower-tail trajectory updating after partial observation of a stay.

Candidates whose validation performance is statistically indistinguishable under a one-standard-error rule are resolved in favor of the simpler configuration, particularly with respect to K. The retained specification is then checked for stability of the fitted basis functions and for convergence behavior across multiple starting values.

3.4 | Empirical uncertainty summaries

The main inferential targets of the proposed model are the common parameter block and the reference conditional quantile trajectory. The heterogeneous component is also of scientific interest, but because the loading matrix is only identified up to orthogonal rotation, it is not natural to base inference on individual entries of C_τ. For this reason, uncertainty assessment is handled differently for the common and heterogeneous parts of the model.

For the common parameter block, we use cluster-level bootstrap resampling. ICU stays are resampled with replacement, and the model is refitted on each bootstrap sample using the same tuning values selected in the original data. This resampling scheme preserves the within-stay dependence structure and provides a pragmatic way to quantify uncertainty for α_τ, β_τ, θ_τ, and functionals of the fitted reference trajectory. Because the present paper does not establish a full asymptotic validity theory for bootstrap inference under bounded cluster sizes and profiled stay-specific scores, these intervals are reported as pragmatic uncertainty summaries rather than exact finite-sample guarantees. Their empirical interval behavior is therefore examined directly in the simulation study through coverage and interval-width summaries.

For the fitted reference trajectory

$$\hat{q}_\tau^{\mathrm{ref}}(t) = \hat{\alpha}_\tau + \widetilde{B}(t)^\top \hat{\theta}_\tau,$$

the bootstrap is used to construct pointwise intervals over a clinically relevant time grid. For the heterogeneous component, we do not report entrywise significance tests for the loading matrix. Instead, we summarize uncertainty through rotation-invariant objects such as the stability of the estimated subspace across bootstrap samples, the shape of aligned deviation functions, and reconstructed lower-tail trajectories for representative stays.

4 | STATISTICAL TARGET AND LIMITED LARGE-SAMPLE JUSTIFICATION

Because the stay-specific scores are profiled nuisance parameters and cluster sizes are bounded in the theoretical approximation, we formulate the large-sample target in terms of the smoothed and penalized criterion actually implemented in practice. Throughout this section, the smoothing parameter $h > 0$, the score penalty $\lambda_a > 0$, the loading penalty $\lambda_R \geq 0$, the spline dimension L, and the working rank K are treated as fixed. The target is therefore the pseudo-true minimizer of the implemented smoothed and penalized criterion, not the unsmoothed structural conditional quantile model. Statements connecting this target to an unsmoothed structural quantile curve would require additional tuning-sequence assumptions, such as $h = h_N \downarrow 0$ and suitably controlled penalty sequences; those assumptions are not imposed here.

For a stay-level observation

$$W_i = \{Y_{ij}, t_{ij}, x_i^\circ : j = 1, \dots, n_i\},$$

define the profiled stay-level loss

$$m_{\tau,h,\lambda}(W_i; \vartheta_\tau, C_\tau) = \min_{a \in \mathbb{R}^K} \left[\sum_{j=1}^{n_i} \rho_{\tau,h} \left\{ Y_{ij} - \alpha_\tau - x_i^{\circ\top} \beta_\tau - \widetilde{B}(t_{ij})^\top \theta_\tau - \widetilde{B}(t_{ij})^\top C_\tau a \right\} \right.$$
$$\left. + \lambda_a \|a\|_2^2 \right].$$

The corresponding population profiled criterion is

$$\mathcal{Q}_{\tau,h,\lambda}(\vartheta_\tau, C_\tau) = E\left\{ m_{\tau,h,\lambda}(W_i; \vartheta_\tau, C_\tau) \right\} + \lambda_R \mathcal{R}(C_\tau),$$

for $(\vartheta_\tau, C_\tau) \in \Theta \times \mathcal{C}_K$. Its empirical profiled version is

$$\mathcal{Q}^p_{N,\tau,h,\lambda}(\vartheta_\tau, C_\tau) = \frac{1}{N} \sum_{i=1}^{N} m_{\tau,h,\lambda}(W_i; \vartheta_\tau, C_\tau) + \lambda_R \mathcal{R}(C_\tau).$$

The algorithm in Section 3 optimizes the equivalent unprofiled version in (7), with the stay-specific scores updated as explicit nuisance parameters. The profiled criterion is used here to define the large-sample target and to state the theoretical approximation.

Let $(\vartheta^\circ_{\tau,h,\lambda}, \mathcal{S}^\circ_{\tau,h,\lambda})$ denote a locally separated minimizer of $\mathcal{Q}_{\tau,h,\lambda}$ in the sense of Assumption 4, where $\mathcal{S}^\circ_{\tau,h,\lambda}$ is the corresponding low-rank subspace. This pseudo-true pair is the theoretical object justified below.

Proposition 2 (Uniform convergence of the profiled criterion)**.** *Under Assumptions 1–3,*

$$\sup_{\vartheta_\tau \in \Theta,\ C_\tau \in \mathcal{C}_K} \left| \mathcal{Q}^p_{N,\tau,h,\lambda}(\vartheta_\tau, C_\tau) - \mathcal{Q}_{\tau,h,\lambda}(\vartheta_\tau, C_\tau) \right| = o_p(1).$$

Proof sketch. The parameter space $\Theta \times \mathcal{C}_K$ is fixed-dimensional and compact. For fixed $h > 0$, the smooth quantile loss $\rho_{\tau,h}$ is Lipschitz in the residual. Together with bounded design, bounded within-stay observation counts, and finite first moments of the responses, this gives an integrable envelope for the stay-level loss class. The condition $\lambda_a > 0$ makes the inner score problem coercive, and Assumption 3 gives measurable profiled losses that are almost surely continuous in (ϑ_τ, C_τ). The profiled stay-level losses therefore form a Glivenko–Cantelli class, yielding the stated uniform law of large numbers.

Proposition 3 (Local consistency for the pseudo-true target)**.** *Suppose Assumptions 1–4 hold. Let $(\hat{\vartheta}_\tau, \widehat{C}_\tau)$ be a sequence of approximate local minimizers of $\mathcal{Q}^p_{N,\tau,h,\lambda}$ in the separated neighborhood of $(\vartheta^\circ_{\tau,h,\lambda}, \mathcal{S}^\circ_{\tau,h,\lambda})$, with optimization error $o_p(1)$ relative to the local empirical minimum. Then*

$$\|\hat{\vartheta}_\tau - \vartheta^\circ_{\tau,h,\lambda}\|_2 + d\left\{ \mathcal{S}(\widehat{C}_\tau), \mathcal{S}^\circ_{\tau,h,\lambda} \right\} = o_p(1).$$

Proof sketch. By Proposition 2, the empirical profiled criterion converges uniformly to the population profiled criterion on the compact local parameter-subspace neighborhood. Assumption 4 gives strict separation of the population criterion on the boundary of any sufficiently small neighborhood of the pseudo-true pair. The standard argmin consistency argument then implies that an approximate empirical local minimizer with $o_p(1)$ optimization error cannot remain outside that neighborhood with nonvanishing probability.

These propositions are intentionally limited in scope. They justify consistency for the pseudo-true target associated with the implemented fixed-tuning estimator. We do not claim here a full structural asymptotic theory for the unsmoothed conditional quantile model under bounded cluster sizes, nor formal bootstrap validity after data-driven tuning selection. Those questions are deferred to future work and are instead examined empirically in the present paper through simulation and sensitivity analyses.

5 | SIMULATION STUDY

The simulation study is designed to assess the finite-sample behavior of the proposed quantile-adaptive profiled-score mixed-effects representation under controlled settings that reflect the main features of irregular ICU monitoring data. The primary goal is not merely to show that the optimization routine can be executed, but to evaluate whether the estimator can recover lower-tail trajectory structure in a statistically meaningful and numerically stable manner. In particular, the simulations complement the modest justification of Section 4 by evaluating how the estimator behaves when observation times are irregular, the heterogeneous trajectory structure must be learned jointly with the common component, and the response distribution may deviate from light-tailed benchmark assumptions.

5.1 | Simulation objectives

The first objective of the simulation study is to evaluate estimation accuracy for the common parameter block. Since the main scientific target of the method is the lower-tail trajectory rather than merely a coefficient vector, we pay particular attention to the fitted curve

$$\hat{q}_\tau^{\mathrm{ref}}(t) = \hat{\alpha}_\tau + \widetilde{B}(t)^\top \hat{\theta}_\tau$$

over the entire observation window.

The second objective is to assess recovery of the heterogeneous low-rank trajectory structure. Because the loading matrix is learned jointly under the quantile objective, this part of the simulation directly evaluates the main methodological contribution of the paper. The relevant question is not only whether the fitted model predicts well, but whether it correctly recovers the dominant directions of lower-tail trajectory heterogeneity and the resulting stay-specific deviation processes.

The third objective is to assess numerical behavior of the blockwise algorithm. Since the estimator is defined through a nonconvex optimization problem, finite-sample performance depends not only on statistical identification but also on convergence stability, sensitivity to initialization, and the interaction between smoothing and regularization. The simulation study therefore records both statistical and computational summaries, allowing us to distinguish between poor recovery due to genuine model difficulty and poor recovery due to numerical instability.

5.2 | Data-generating mechanism

For each simulation replicate, we generate N longitudinal trajectories over the interval $[0, T]$, where $T = 24$. The basic sampling unit is a stay, indexed by $i = 1, \dots, N$, and the repeated measurements within stay i are indexed by $j = 1, \dots, n_i$. The response is generated according to

$$Y_{ij} = \alpha_0 + x_i^{\circ\top} \beta_0 + \widetilde{B}_0(t_{ij})^\top \theta_0 + \widetilde{B}_0(t_{ij})^\top C_0 a_i + \varepsilon_{ij}, \tag{12}$$

where $x_i^\circ \in \mathbb{R}^{p-1}$ is a low-dimensional baseline covariate vector, $\widetilde{B}_0(t)$ is the centered basis system used in data generation, $C_0 \in \mathbb{R}^{L_0 \times K_0}$ is the true loading matrix, and $a_i \in \mathbb{R}^{K_0}$ is the latent score vector. Throughout the main simulations, the target quantile level is fixed at $\tau = 0.10$, reflecting the lower-tail emphasis of the motivating application.

In the baseline setting, we take $p = 2$ and generate $x_i^\circ \sim N(0, 1)$. The true common trajectory is chosen so that the reference lower-tail MAP curve begins at a relatively depressed level, rises during the early hours, and then stabilizes. The heterogeneous component has rank $K_0 = 2$: the first deviation direction represents an approximately global downward or upward shift in the lower-tail trajectory, while the second represents delayed recovery with comparatively lower early values. The score vectors are generated independently and then centered within each replicate so that the finite-sample reference trajectory remains exactly equal to the common component. The error term is generated so that its conditional τ-quantile is zero.

Observation times are generated to mimic irregular ICU monitoring. In the baseline scenario, the number of observations per stay is sampled from a moderate-count distribution such as $\mathrm{Unif}\{8, \dots, 15\}$, and the observation times are obtained by perturbing an approximately regular grid over $[0, 24]$. This produces trajectories that are irregular but not severely sparse.

5.3 | Simulation scenarios

We consider eight scenarios. Scenario A is the baseline feasibility setting, in which the data are generated exactly from the model in (12). Scenario B increases the irregularity of the observation-time design by allowing more variable stay-level counts and more uneven timing over the 24-hour window. Scenario C replaces the baseline light-tailed error by a heavier-tailed alternative while preserving the condition that the conditional τ-quantile is zero. Scenario D replaces the Gaussian score distribution by a non-Gaussian alternative, such as a centered multivariate *t* distribution or an asymmetric finite mixture. Scenario E examines basis misspecification by fitting the model with a working basis dimension smaller or larger than the basis dimension used to generate the data. Scenario F investigates sensitivity to initialization by comparing the structured low-rank start with random perturbation starts. Scenario G introduces heteroscedasticity, allowing the error scale to vary with the score vector and time so that lower-tail geometry differs from mean-oriented geometry. Scenario H allows the observation process to be informative, in the sense that low recent MAP values increase the chance of obtaining an additional measurement in the next interval. Scenarios A, C, G, and H are the primary scenarios for the first report: Scenario A checks baseline recovery, Scenario C tests heavy-tailed noise, Scenario G is the key setting for assessing whether lower-tail geometry differs from mean-oriented geometry, and Scenario H evaluates sensitivity to informative observation.

Across all scenarios, we consider at least two sample-size settings, for example $N = 100$ and $N = 200$, and use `[SIM_REPLICATES]` Monte Carlo replicates per configuration. The exact coefficient values, basis coefficients, and scenario-specific distributions are recorded in the supplementary simulation code.

5.4 | Competing methods

To isolate the contribution of quantile-adaptive basis learning, we compare the proposed method with four practical benchmarks. The first comparator is a marginal spline-based quantile trajectory model with $K = 0$, which omits the stay-specific low-rank deviation term altogether. The second comparator is a random intercept-slope longitudinal quantile mixed model, representing a simpler and clinically familiar alternative. The core comparator is a two-stage FPCA-plus-quantile procedure in which the trajectory basis is estimated under a mean-oriented FPCA criterion and then held fixed in a downstream quantile fit. The fourth comparator is a mean-based reduced-rank mixed model fitted by squared error. These methods distinguish the value of explicit heterogeneity modeling, the value of quantile-adaptive basis learning, and the value of moving beyond conventional random intercept-slope quantile models. All methods are evaluated using the same training and validation splits within each Monte Carlo replicate.

5.5 | Performance measures

The finite-sample performance of the competing methods is evaluated from four complementary perspectives. The first set of measures concerns estimation accuracy for the common parameters, with particular emphasis on the integrated squared error

$$\mathrm{ISE}(\hat{q}_\tau^{\mathrm{ref}}) = \int_0^T \left\{ \hat{q}_\tau^{\mathrm{ref}}(t) - q_0^{\mathrm{ref}}(t) \right\}^2 dt, \tag{13}$$

approximated numerically over a dense time grid. The second set concerns recovery of the heterogeneous component. Because the loading matrix is identifiable only up to orthogonal transformation, entrywise error in $\hat{C}_\tau$ is not appropriate; instead, we record projector-based subspace distance, score correlation after alignment, and the mean integrated squared error of the reconstructed deviation process. The third set concerns prediction and uncertainty summaries, including within-stay forecasting check loss, empirical coverage and mean width of cluster-bootstrap intervals for selected common parameters and time-grid values of $q_{\tau}^{\mathrm{ref}}(t)$, and simulation-based coverage assessment. The fourth concerns numerical behavior, including convergence failure rate, number of iterations, final objective dispersion across starts, and runtime. The finalized simulation report will also include sensitivity to h, K, λ_a, and λ_R.

5.6 | Finite-sample results

Table 1 and Table 2 are the recommended main displays for the simulation study. They are designed to report lower-tail trajectory recovery, heterogeneous subspace recovery, empirical bootstrap interval behavior, and convergence summaries. In the finalized manuscript, the narrative accompanying these tables should address three questions directly: whether the proposed method can improve recovery of the lower-tail trajectory relative to the comparators, whether any improvement is concentrated in Scenario G where lower-tail and mean-based structure differ, and whether the algorithm remains numerically stable under irregular or informative observation designs.

TABLE 1 Recommended main simulation summary for the baseline setting (Scenario A). Replace the placeholders with Monte Carlo averages and Monte Carlo standard errors from the finalized simulation runs. Coverage entries should be reported only for methods with a directly implemented interval procedure; otherwise use — in that column.

Method	ISE of $\hat{q}_{0.10}^{\mathrm{ref}}$	Subspace distance	Coverage (%)	Convergence (%)
Proposed	[simA_prop_ise]	[simA_prop_subspace]	[simA_prop_cov]	[simA_prop_conv]
Marginal quantile	[simA_marg_ise]	—	[simA_marg_cov]	[simA_marg_conv]
RI–RS quantile mixed model	[simA_ri_ise]	—	[simA_ri_cov]	[simA_ri_conv]
Two-stage FPCA-plus-quantile	[simA_2stage_ise]	[simA_2stage_subspace]	[simA_2stage_cov]	[simA_2stage_conv]
Mean-based reduced-rank model	[simA_mean_ise]	[simA_mean_subspace]	[simA_mean_cov]	[simA_mean_conv]

TABLE 2 Recommended robustness summary across Scenarios B–H. Replace the placeholders with the finalized simulation summaries. The coverage range should summarize only those methods for which a directly comparable interval procedure is implemented in the finalized experiments.

Scenario	Best method by ISE	Coverage range (%)	Key numerical note
B: stronger irregularity	[simB_best]	[simB_covrange]	[simB_note]
C: heavy-tailed error	[simC_best]	[simC_covrange]	[simC_note]
D: non-Gaussian scores	[simD_best]	[simD_covrange]	[simD_note]
E: basis misspecification	[simE_best]	[simE_covrange]	[simE_note]
F: initialization sensitivity	[simF_best]	[simF_covrange]	[simF_note]
G: heteroscedastic lower-tail geometry	[simG_best]	[simG_covrange]	[simG_note]
H: informative observation process	[simH_best]	[simH_covrange]	[simH_note]

For clarity of interpretation, we recommend summarizing the finalized simulation results in the following order. First, report the baseline finite-sample behavior in Scenario A, emphasizing the accuracy of the reference lower-tail trajectory, projector-based recovery of the heterogeneous component, and empirical bootstrap interval behavior. Second, discuss the robustness scenarios, with special attention to Scenario G, where differences between quantile-adaptive and mean-oriented comparators are expected to be most informative if lower-tail geometry genuinely differs from mean-oriented geometry. Third, report convergence failure rate, objective-value stability across starts, and any cases in which the structured multi-start routine was necessary to avoid poor local minima.

6 | APPLICATION TO THE MIMIC-IV DATASET

6.1 | Data source and study cohort

The application is designed to use ICU monitoring data extracted from MIMIC-IV, a large publicly available critical care electronic health record database [1,2]. The finalized extraction will report the exact MIMIC-IV version, `[MIMIC_VERSION]`, used for cohort construction. The empirical analysis focuses on early mean arterial pressure trajectories during ICU stay. Consistent with the modeling framework developed above, the ICU stay is treated as the primary unit of longitudinal analysis, and time is aligned relative to ICU admission.

The study cohort is defined to reflect the clinical and statistical aims of the paper. The primary analysis restricts the sample to adult ICU stays (age $\geq$ 18 years) with available MAP measurements during the first 24 hours after ICU admission, after applying the inclusion and exclusion criteria documented in Table 3. To preserve the stay-level independence assumption used throughout the paper, the primary analysis retains the first eligible ICU stay for each subject; a sensitivity analysis may relax this rule and cluster by subject. Stays are included if they have at least `[MIN_MAP_COUNT]` MAP measurements within the analysis window. MAP is extracted from item identifiers `[ITEM_ID_LIST]`. Physiologically implausible values below `[LOW_MAP]` mmHg or above `[HIGH_MAP]` mmHg are removed. Measurements recorded within the same minute are aggregated according to the rule `[DUPLICATE_RULE]`, and when invasive and noninvasive MAP measurements are concurrent, the priority rule `[INVASIVE_RULE]` is applied. After preprocessing, the finalized analysis will report the analytic cohort size, `[N_stays]` ICU stays from `[N_subjects]` unique patients, and `[N_MAP]` MAP observations.

This study is descriptive rather than causal. The goal is to summarize lower-tail MAP trajectories during early ICU time and to identify dominant forms of between-stay lower-tail heterogeneity under routine care. Dynamic treatments such as vasopressors, fluids, sedation, and ventilation undoubtedly influence MAP, and the fitted trajectories should not be interpreted as treatment-free physiologic processes or causal effects of baseline characteristics.

6.2 | Outcome definition, covariates, and preprocessing

Let Y_{ij} denote the observed MAP value for ICU stay i at relative time t_{ij}, measured in hours since ICU admission. The primary analysis window was restricted to the first 24 hours of ICU stay, reflecting the early phase of critical care during which hemodynamic instability and active clinical stabilization are especially important.

All measurements were aligned relative to ICU admission time. This choice is appropriate for MIMIC-IV, where subject-level date shifting preserves within-subject temporal ordering but makes direct calendar-time comparisons across subjects inappropriate [1]. The use of ICU-relative time therefore provides a coherent longitudinal timescale for comparing early hemodynamic trajectories across stays.

Baseline covariates included age, sex, admission type, and `[BASELINE_ACUITY_VARIABLE]`. Continuous covariates were standardized to mean zero and variance one before model fitting. Missing baseline covariates were handled by `[MISSINGNESS_RULE]`, and the same rule was used across all models to ensure comparability. The empirical distribution of observation counts per stay and the time-density distribution of measurements over the first 24 hours are reported in Table 3 and Figure 1. These summaries are important because the proposed estimator is designed precisely for settings in which trajectories are unevenly sampled and between-stay heterogeneity cannot be reduced to a balanced repeated-measures structure.

6.3 | Model fitting and comparator models

The proposed quantile-adaptive profiled-score mixed-effects representation will be fitted at the lower quantile level $\tau = 0.10$. The temporal dictionary $\widetilde{B}(t)$ is specified on $[0, 24]$ using a moderate-dimensional spline basis with `[BASIS_DIMENSION]` basis functions before centering. The working rank is selected from `[K_GRID]`. The final tuning configuration will be reported as

$$K = \texttt{[K\_FINAL]}, \qquad h = \texttt{[H\_FINAL]}, \qquad \lambda_a = \texttt{[LAMBDA\_A\_FINAL]}, \qquad \lambda_R = \texttt{[LAMBDA\_R\_FINAL]}.$$

These quantities are selected by the within-stay forecasting scheme described in Section 3.

To assess practical value rather than internal fit alone, the MIMIC-IV application will compare the proposed model with four benchmarks: a marginal spline-based quantile model ($K = 0$), a random intercept-slope quantile mixed model, a two-stage FPCA-plus-quantile procedure using a mean-oriented basis, and a mean-based reduced-rank mixed model. Model comparison uses the same within-stay forecasting protocol used in tuning selection: common parameters are estimated on training stays, stay-specific scores for held-out stays are estimated from early observations only, and predictive performance is evaluated by check loss on later observations at $\tau = 0.10$.

Because the proposed estimator is nonconvex, empirical convergence diagnostics are substantively important rather than merely technical. The finalized analysis should therefore report the number of successful starts, the spread of final objective values, and the stability of the estimated subspace under random perturbation starts. Those summaries are displayed in Supplementary Table S`[STABILITY_TABLE]`.

6.4 | Empirical summaries of interest

The empirical analysis is designed to produce five complementary summaries. The first is the fitted reference lower quantile trajectory

$$\hat{q}_\tau^{\mathrm{ref}}(t) = \hat{\alpha}_\tau + \widetilde{B}(t)^\top \hat{\theta}_\tau,$$

which provides a direct description of how the lower tail of MAP evolves during the first 24 ICU hours for a reference covariate profile. This curve is the primary population-level object of interest.

The second summary is the collection of aligned quantile-adaptive basis functions

$$\hat{\psi}_{k,\tau}(t) = \widetilde{B}(t)^\top \hat{c}_{k,\tau}, \qquad k = 1, \ldots, K.$$

These functions describe the dominant directions of lower-tail trajectory heterogeneity captured by the model. Their shape indicates whether between-stay variation is driven mainly by globally depressed lower-tail MAP, delayed hemodynamic recovery, early transient suppression, or other interpretable forms of deviation.

The third summary is the stay-specific deviation structure represented by the reconstructed trajectories

$$\hat{\delta}_{i,\tau}(t) = \widetilde{B}(t)^\top \hat{C}_\tau \hat{a}_i.$$

These quantities allow representative lower-tail trajectories to be reconstructed and compared across stays. The emphasis is not on the numerical score vectors themselves, which depend on the chosen orientation convention, but on the trajectory patterns generated by the aligned basis functions and fitted scores.

The fourth summary is the comparator-based validation performance. This will evaluate whether the proposed method can improve lower-tail forecasting once partial trajectory information is available for a stay. The fifth summary connects the fitted low-rank structure back to a clinically familiar threshold by examining MAP below 65 mmHg burden, such as observed time below 65 mmHg or the observed proportion of recorded measurements below 65 mmHg, across strata of the dominant score. As optional secondary descriptive evidence, the analysis may report associations between estimated lower-tail score strata and acute kidney injury, ICU mortality, hospital mortality, or ICU length of stay. These clinical outcome summaries are descriptive and are not interpreted as causal effects.

6.5 | Results presentation

Table 3, Figure 1, Figure 2, Figure 3, and Table 4 are the recommended main displays for the MIMIC-IV section. The present draft leaves the analysis-dependent quantities as structured placeholders rather than fabricating numerical values. Once the finalized extraction and fitting pipeline is run, the empirical narrative should be written around the following sequence.

First, report the analytic cohort size, the total number of MAP observations, and the distribution of measurement counts per stay. Second, interpret the fitted reference $\tau = 0.10$ MAP trajectory, focusing on whether the lower-tail curve suggests early instability followed by stabilization or persistent depression over clinically relevant portions of the 24-hour window. Third, describe the aligned deviation directions in clinically meaningful terms, such as global lower-tail depression or delayed recovery, while making clear that these are oriented representatives of an identified subspace rather than uniquely identified basis vectors.

Fourth, compare validation check loss across the proposed model and the benchmark procedures. Finally, relate the dominant score to the observed burden of MAP below 65 mmHg.

TABLE 3 Recommended cohort description table for the MIMIC-IV application. Replace the placeholders with the finalized extraction summaries.

Quantity	Value
MIMIC-IV version	[MIMIC_VERSION]
Eligible adult ICU stays before exclusions	[N_eligible]
Adult inclusion and exclusion criteria	[ADULT_ICU_CRITERIA]
Analytic ICU stays	[N_stays]
Unique subjects	[N_subjects]
First-stay or multiple-stay rule	[STAY_SELECTION_RULE]
Total MAP observations	[N_MAP]
Median MAP observations per stay (IQR)	[MEDIAN_COUNT_IQR]
Minimum observation-count criterion	[MIN_MAP_COUNT]
Item identifiers used for MAP extraction	[ITEM_ID_LIST]
Physiologic MAP filter	[LOW_MAP]–[HIGH_MAP] mmHg
Duplicate handling rule	[DUPLICATE_RULE]
Concurrent invasive/noninvasive priority rule	[INVASIVE_RULE]
Baseline acuity variable	[BASELINE_ACUITY_VARIABLE]
Time-varying treatment sensitivity variables	[TREATMENT_VARIABLES]
Threshold-based hypotension summary	[MAP65_SUMMARY]
Secondary descriptive outcomes	[SECONDARY_OUTCOMES]

FIGURE 1 Placeholder for the empirical distribution of measurement density over the first 24 hours after ICU admission. Replace the box with the finalized figure file.

6.6 | Sensitivity analyses

Sensitivity analyses are important because the empirical findings depend on both modeling and preprocessing choices. The most relevant sensitivity checks concern the working rank K, the smoothing parameter h, the basis dimension, the handling of duplicate timestamps, and the observation-process assumption. The first sensitivity analysis varies K over a small range of plausible values to examine whether the main lower-tail trajectory pattern and the dominant heterogeneous directions remain stable when the complexity of the low-rank component is slightly increased or decreased. The second varies the smoothing parameter and nearby regularization values to verify that the fitted lower-tail trajectory and the learned deviation structure do not change materially under modest perturbations of the numerical smoothing regime. The third examines alternative preprocessing rules, including different handling of duplicate timestamps, alternative plausibility filters, and different minimum observation-count thresholds.

FIGURE 2 Placeholder for the fitted reference $\tau = 0.10$ MAP trajectory with pointwise cluster-bootstrap intervals. Replace the box with the finalized figure file.

FIGURE 3 Placeholder for aligned quantile-adaptive deviation directions and representative reconstructed lower-tail trajectories. Replace the box with the finalized figure file.

TABLE 4 Recommended validation comparison for the MIMIC-IV application. Replace the placeholders with within-stay forecasting summaries from the finalized analysis.

Method	Validation check loss at $\tau = 0.10$
Proposed quantile-adaptive basis model	`[prop_checkloss]`
Marginal spline quantile model	`[marg_checkloss]`
Random intercept-slope quantile mixed model	`[ri_checkloss]`
Two-stage FPCA-plus-quantile model	`[twostage_checkloss]`
Mean-based reduced-rank mixed model	`[mean_checkloss]`

Because ICU measurement frequency may depend on patient acuity, we also fit an inverse-intensity weighted sensitivity analysis. Let $\hat{\pi}_{ij}$ denote the fitted probability that stay i is observed in interval j, estimated from a pooled logistic model using time since ICU admission, baseline covariates, and the most recent available MAP value. The weighted fit replaces $\sum_{ij} \rho_{\tau,h}(r_{ij})$ by $\sum_{ij} \hat{\pi}_{ij}^{-1} \rho_{\tau,h}(r_{ij})$. Similar findings in this analysis would support the stability of the primary descriptive results to departures from conditional exogeneity of the observation process.

As a further descriptive sensitivity analysis, the model may be augmented with time-varying indicators of vasopressor exposure, fluid administration, invasive mechanical ventilation, and sedation:

$$Q_\tau(Y_{ij} \mid x_i^\circ, z_{ij}, t_{ij}, a_i) = \alpha_\tau + x_i^{\circ\top} \beta_\tau + z_{ij}^\top \gamma_\tau + \widetilde{B}(t_{ij})^\top \theta_\tau + \widetilde{B}(t_{ij})^\top C_\tau a_i,$$

where z_{ij} collects the time-varying treatment indicators. This extension remains descriptive rather than causal; its purpose is to assess whether the principal lower-tail trajectory patterns persist after adjustment for major concurrent interventions.

7 DISCUSSION

In this paper, we have developed a quantile-adaptive profiled-score mixed-effects representation for irregular longitudinal trajectories, motivated by the need to characterize lower-tail hemodynamic behavior in ICU monitoring data. The central idea is to replace the conventional two-stage strategy of first learning a low-dimensional trajectory basis under a mean-oriented criterion and then fitting a downstream quantile model with a joint formulation in which the low-rank trajectory representation is learned directly under the target quantile objective. This construction yields a model that combines three desirable features within a single framework: a flexible reference conditional quantile trajectory, a parsimonious representation of stay-specific heterogeneity, and an estimation procedure that remains feasible in the presence of irregular observation times.

A major benefit of the revised formulation is that the main non-identifiabilities are handled explicitly. The intercept is separated from the time-varying component through a centered basis dictionary, and the translational non-identifiability between the common trajectory and the stay-specific scores is removed by the empirical centering constraint on the score vectors. The resulting working representation supports interpretable decomposition of lower-tail MAP dynamics into a reference conditional trajectory and a small number of aligned heterogeneous deviation patterns, while retaining the fact that the loading matrix is identified only up to rotation.

A second contribution is that the paper now treats basis learning, estimation, and uncertainty assessment as parts of a coherent modeling strategy rather than as disconnected steps. The proposed estimator is built from a smooth quantile objective with regularization on both the stay-specific scores and the loading matrix, and it is implemented through a blockwise algorithm that alternates among common parameters, score updates, and the low-rank trajectory representation. The validation scheme is designed around within-stay forecasting, thereby avoiding the conceptual ambiguity that arises when one evaluates a held-out stay without specifying how its heterogeneous score is to be estimated.

The present method should be interpreted as a descriptive and partially predictive model for lower-tail MAP evolution after partial observation of a stay. Its main practical value is to summarize cohort-level lower-tail dynamics and to decompose between-stay variation into a small number of clinically interpretable trajectory patterns. It is not a causal model of treatment effects, and the estimated deviation directions are best interpreted as low-dimensional summaries of observed trajectory heterogeneity under routine ICU care. The stay-specific score a_i is a profiled score, not a structural random effect drawn from and integrated over a parametric mixing distribution. For a completely new stay with no within-stay MAP observations, the model provides only the reference trajectory together with the shift implied by baseline covariates. Stay-specific forecasting becomes available only after partial observations have been used to estimate the nuisance score vector a_i.

The theoretical development in this paper is deliberately modest. Rather than asserting a full structural asymptotic theory that would require much stronger assumptions and considerably more technical work, we justify the estimator relative to the pseudo-true target defined by the smoothed and penalized criterion actually optimized in practice. For a journal such as *Statistics in Medicine*, this balance is appropriate: the key requirement is not maximal technical generality, but a method that is statistically coherent, empirically interpretable, and usable in a realistic biomedical setting.

Several limitations remain. First, the heterogeneous component is naturally summarized through its estimated subspace and aligned deviation functions rather than through entrywise inference for the loading matrix. This is not a weakness of the analysis so much as a consequence of the identification structure of the model: the loading matrix is only defined up to orthogonal transformation, and the clinically meaningful object is the induced deviation process rather than a particular rotated representative. Second, the empirical application is observational and descriptive. Dynamic treatments such as vasopressors, fluids, ventilation, and sedation all influence MAP trajectories, so the fitted lower-tail patterns should not be interpreted causally. Third, the observation process may be informative in ICU data because unstable patients are often measured more frequently. We therefore treat conditional exogeneity as a working assumption and examine its impact through inverse-intensity weighted sensitivity analysis. Fourth, bootstrap intervals are pragmatic uncertainty summaries whose empirical interval behavior is examined in simulation; the paper does not establish formal bootstrap validity. Fifth, for fixed h the theoretical target is a smoothed pseudo-true target, and the current theory does not cover consistency of data-driven tuning selection or recovery of an unsmoothed structural target as $h \to 0$.

These limitations point directly to several useful extensions. One is to allow time-varying treatment indicators or intervention processes to enter the model more systematically, thereby making it possible to study how vasopressor exposure, fluid administration, or other dynamic clinical factors co-evolve with lower-tail hemodynamic trajectories. Another is to extend the quantile-adaptive basis idea to multivariate physiologic streams, since clinically important deterioration often manifests jointly across blood pressure, lactate, urine output, and other monitored variables. A third direction is to develop sharper inferential theory for the heterogeneous component itself, including uncertainty quantification for aligned deviation functions and reconstructed stay-specific lower-tail trajectories.

More broadly, the contribution of the present paper is to show that lower-tail trajectory analysis in irregular longitudinal data can be approached through a model that is both structured and distribution-sensitive. Standard reduced-rank methods remain valuable when mean and covariance structure are the primary objects of interest, but they are not automatically aligned with clinically salient lower-tail behavior. The proposed quantile-adaptive profiled-score mixed-effects representation addresses this mismatch directly by bringing basis learning into the quantile objective itself. For ICU monitoring data, where clinical concern is often driven by extremes rather than averages, such alignment is especially important.

□

APPENDIX

A IMPLEMENTATION DETAILS

The estimation algorithm described in Section 3 was implemented in Python using `NumPy` and `SciPy`. The dictionary $B^{\star}(t)$ was constructed as a cubic B-spline basis on $[0, T]$; the centered dictionary $\widetilde{B}(t)$ was obtained by subtracting the time average of the basis vector over the analysis window. The Gram matrix $G_{\widetilde{B}}$ was computed numerically on a dense regular time grid over $[0, T]$, and the same grid geometry was used for weighted orthonormalization of the loading matrix.

Smoothing of the quantile loss employed the same logistic surrogate as in Section 3,

$$\rho_{\tau,h}(r) = \tau r + h \log\{1 + \exp(-r/h)\}.$$

Initialization combines an initial marginal quantile fit for $(\alpha_\tau, \beta_\tau, \theta_\tau)$ with ridge-stabilized stay-level residual basis coefficients, followed by a truncated singular value decomposition of the whitened coefficient matrix. For each reported fit, the planned implementation uses one structured initialization and 20 random perturbation starts. The feasible solution with the smallest final objective value is retained. After each full sweep, the score mean is reabsorbed into θ_τ according to the identifying convention (3), and the loading matrix is retracted to the constraint set through a polar-decomposition update under the $G_{\widetilde{B}}$ metric. When a full sweep increases the objective, a safeguarded damping step blends the previous and current iterates and retains the best feasible nonincreasing candidate. Convergence is declared when either the maximum parameter change falls below 10^{-5} or the objective has stabilized over a short run of consecutive iterations.

All simulation experiments will use `[SIM_REPLICATES]` replicates per scenario. The cluster bootstrap for the MIMIC-IV application will use `[N_BOOT]` resampled datasets. A blinded reproducibility bundle containing SQL extraction code, MAP preprocessing scripts, simulation code, and model-fitting functions should be supplied with the revision rather than deferred to post-acceptance release.

REFERENCES

1. Johnson AEW, Bulgarelli L, Shen L, et al. MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data.* 2023;10:1. doi: 10.1038/s41597-022-01899-x
2. Johnson A, Bulgarelli L, Pollard T, et al. MIMIC-IV (version 3.1). PhysioNet; 2024
3. Evans L, Rhodes A, Alhazzani W, others . Surviving Sepsis Campaign: International Guidelines for Management of Sepsis and Septic Shock 2021. *Intensive Care Medicine.* 2021;47(11):1181–1247. doi: 10.1007/s00134-021-06506-y
4. Schuurmans J, van Rossem BTB, Rellum SR, et al. Hypotension during intensive care stay and mortality and morbidity: a systematic review and meta-analysis. *Intensive Care Medicine.* 2024;50(4):516–525. doi: 10.1007/s00134-023-07304-4
5. Koenker R, Bassett GJ. Regression Quantiles. *Econometrica.* 1978;46(1):33–50.
6. Koenker R. Quantile Regression for Longitudinal Data. *Journal of Multivariate Analysis.* 2004;91(1):74–89. doi: 10.1016/j.jmva.2004.05.006
7. Geraci M, Bottai M. Quantile Regression for Longitudinal Data Using the Asymmetric Laplace Distribution. *Biostatistics.* 2007;8(1):140–154. doi: 10.1093/biostatistics/kxj039
8. Liu Y, Bottai M. Mixed-Effects Models for Conditional Quantiles with Longitudinal Data. *The International Journal of Biostatistics.* 2009;5(1). Article 28doi: 10.2202/1557-4679.1186
9. Geraci M. Linear Quantile Mixed Models: The lqmm Package for Laplace Quantile Regression. *Journal of Statistical Software.* 2014;57(13):1–29. doi: 10.18637/jss.v057.i13
10. Li H, Liu Y, Luo Y. Double Penalized Quantile Regression for the Linear Mixed Effects Model. *Journal of Systems Science and Complexity.* 2020;33:2080–2102.
11. Yao F, Müller HG, Wang JL. Functional Data Analysis for Sparse Longitudinal Data. *Journal of the American Statistical Association.* 2005;100(471):577–590. doi: 10.1198/016214504000001745

12. Morris JS, Carroll RJ. Wavelet-based Functional Mixed Models. *Journal of the Royal Statistical Society, Series B.* 2006;68(2):179–199. doi: 10.1111/j.1467-9868.2006.00539.x
13. Zhou L, Huang JZ, Martinez JG, Maity A, Baladandayuthapani V, Carroll RJ. Reduced Rank Mixed Effects Models for Spatially Correlated Hierarchical Functional Data. *Journal of the American Statistical Association.* 2010;105(489):390–400. doi: 10.1198/jasa.2010.tm08737
14. Scheipl F, Staicu AM, Greven S. Functional Additive Mixed Models. *Journal of Computational and Graphical Statistics.* 2015;24(2):477–501. doi: 10.1080/10618600.2014.901914
15. Kato K. Estimation in Functional Linear Quantile Regression. *The Annals of Statistics.* 2012;40(6):3108–3136. doi: 10.1214/12-AOS1066
16. Li M, Wang K, Maity A, Staicu AM. Inference in Functional Linear Quantile Regression. *Journal of Multivariate Analysis.* 2022;190:104985. doi: 10.1016/j.jmva.2022.104985
17. Méndez-Civieta Á, Wei Y, Diaz KM, Goldsmith J. Functional Quantile Principal Component Analysis. *Biostatistics.* 2025;26(1):kxae040. doi: 10.1093/biostatistics/kxae040
18. Asfar P, Meziani F, Hamel JF, others . High versus Low Blood-Pressure Target in Patients with Septic Shock. *New England Journal of Medicine.* 2014;370:1583–1593. doi: 10.1056/NEJMoa1312173